pfi

Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Scaling and K-Means

WEEK 3 · DAY 2

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From a target to no target

Days 1 to 11: a target

A spread, a default flag, a yield. A function fitted to the target, judged by its errors on rows it never saw; on dated rows, later rows.

Today: no target

Groups of similar bonds across four columns at once: log OAS, duration, coupon, rating score. 821 bonds, 150 invented issuers.

Synthetic data: a cluster here is a group of similar rows in an invented file, and nothing more.

Learning without a target

	Days 1 to 11	Today
Fitted to	A target column	The columns themselves
Returns	A fitted value or probability	A cluster number per bond
Judged by	Errors on test rows	Tightness, separation, judgement
Rows put away	Test rows, first	None: nothing held out

Scaling on all bonds is not leakage today: there are no test rows. As a model feature, it would go inside the pipeline.

How Euclidean distance works

The distance	The difference in each column, squared, added up, then the square root
Units	The formula never sees them: bp, years, and notches count by the size of their numbers
The risk	A column with big numbers can drown the others
The fix	A z-score rewrites each column in standard deviations
What comes out	One distance for each pair of rows, small for rows that are alike

Distance, and why units decide it

```
distance = sqrt( (difference in col 1)^2 + (difference in col 2)^2 + ... )
```

In the file's units: distance 39.02

Column	Difference	Share (%)
oas	39.000	99.90
duration	0.834	0.05
coupon	0.875	0.05
score	0.000	0.00

In standard deviations: distance 0.6244

Column	Difference (sd)	Share (%)
oas	0.210	11.35
duration	0.257	16.88
coupon	0.529	71.77
score	0.000	0.00

1 bp counts as much as 1 year: unscaled, the biggest unit decides the distance.

How K-means works

Start	k centres, with no target to guide the groups
Assign, update	Each row joins its nearest centre; each centre moves to the average of its cluster; repeat until none moves
Inertia	The sum of squared distances to each row's own centre: it never rises, so the loop ends
The starts	k-means++ spreads the starting centres out; n_init fits are run, and the lowest inertia is kept
What comes out	A cluster number for each row, the centres, and the inertia

K-means, step by step

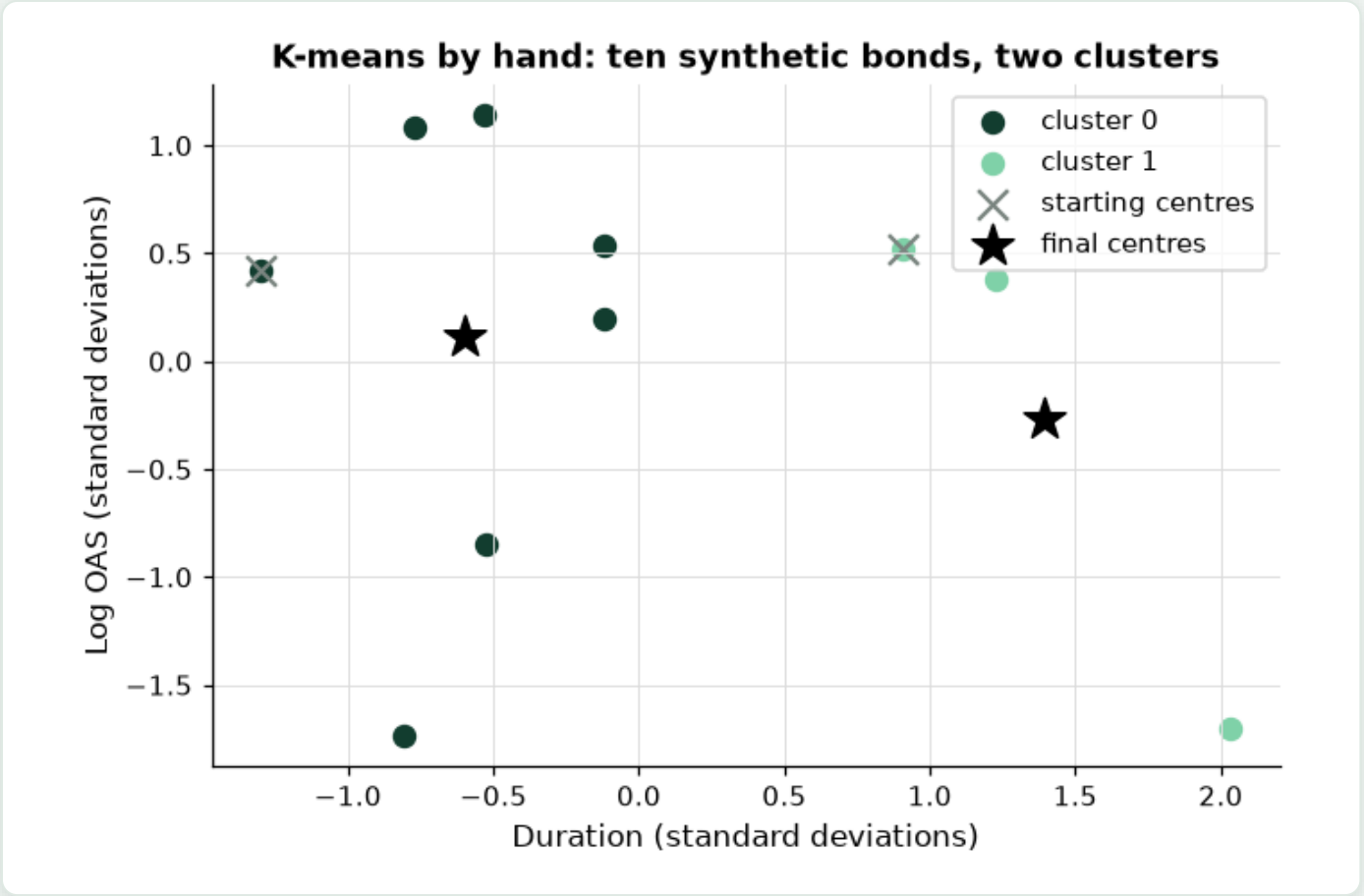
1. Start — k and k starting centres

2. Assign — each bond to nearest centre

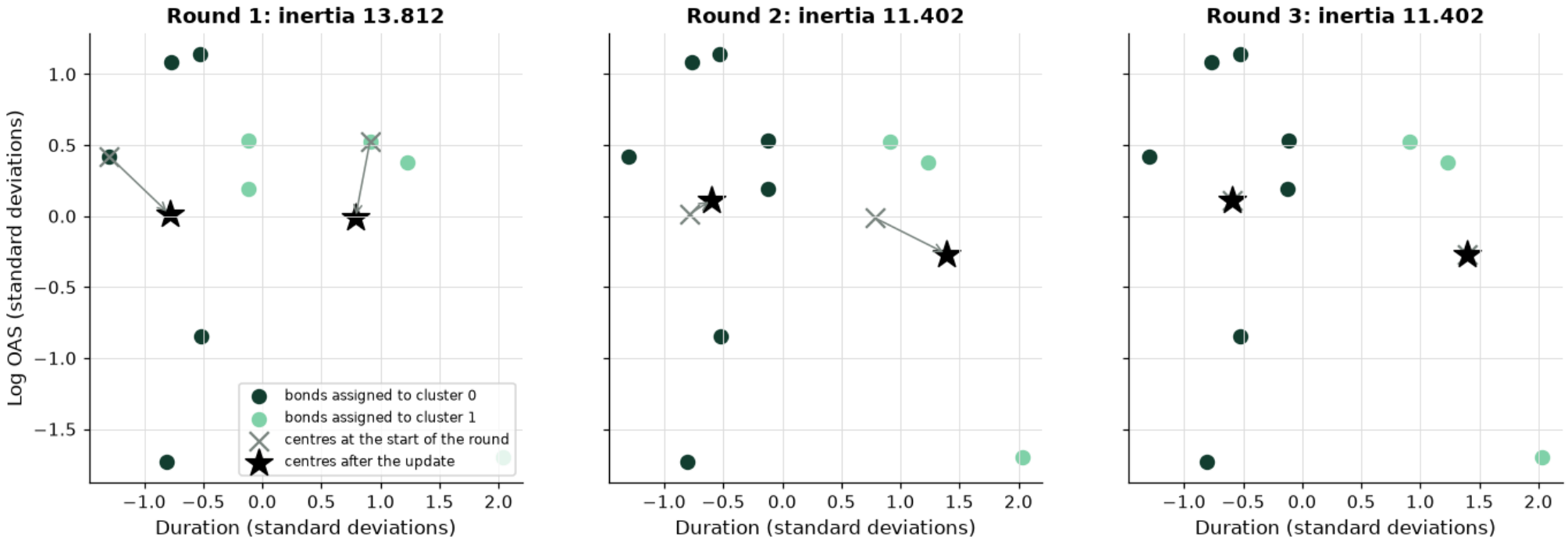
3. Update — centre = its bonds' average

4. Repeat — until no centre moves

Round	Sizes	Inertia
1	5 and 5	13.812
2	7 and 3	11.402
3	7 and 3	11.402

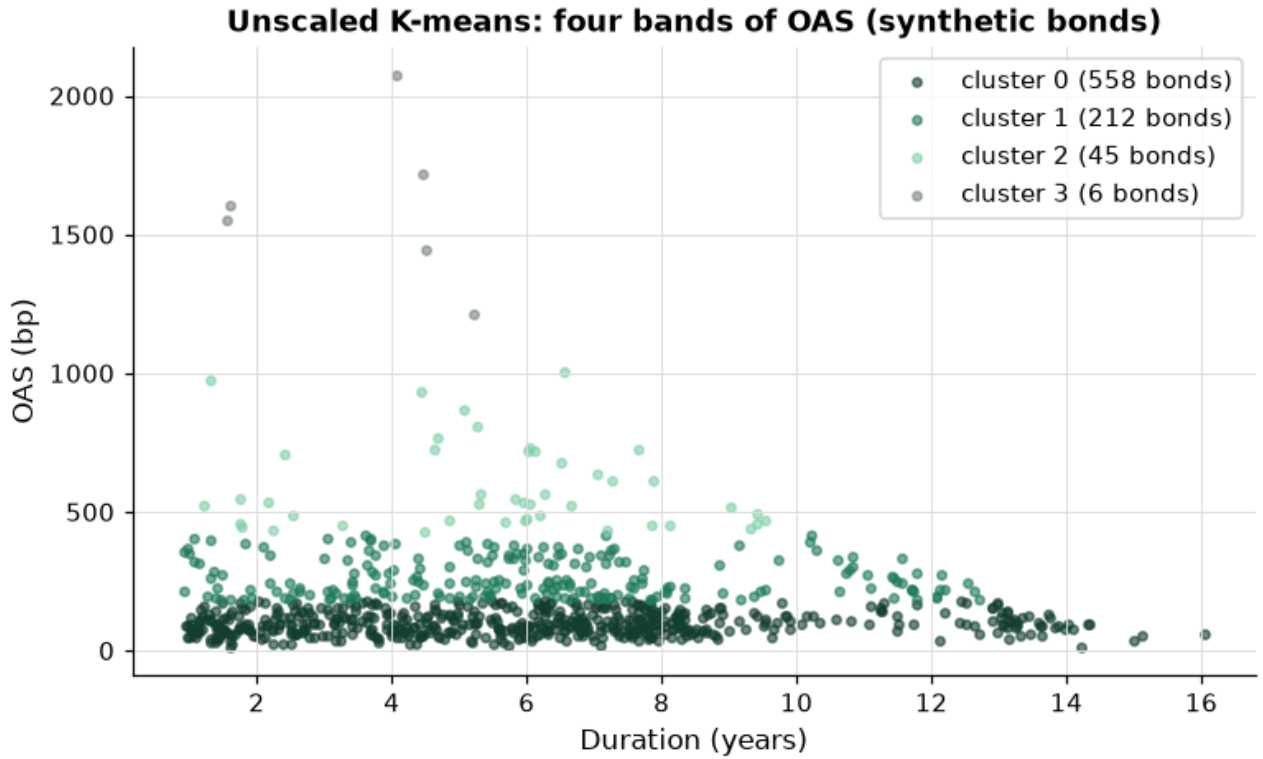


K-means in rounds



Inertia 13.812 after round 1, 11.402 after round 2, and 11.402 after round 3, when nothing moves and the loop stops.

Unscaled: four bands of OAS



How clusters are numbered

By size, largest first: cluster 0 is the largest. A number names a group, not a score.

Cluster	Bonds	OAS (bp)	Duration
0	558	15 to 181	0.9 to 16.0
1	212	182 to 418	0.9 to 12.7
2	45	427 to 1004	1.2 to 9.5
3	6	1215 to 2074	1.5 to 5.2

Cut by OAS alone: not duration groups, not rating groups.

Before and after scaling

10.22

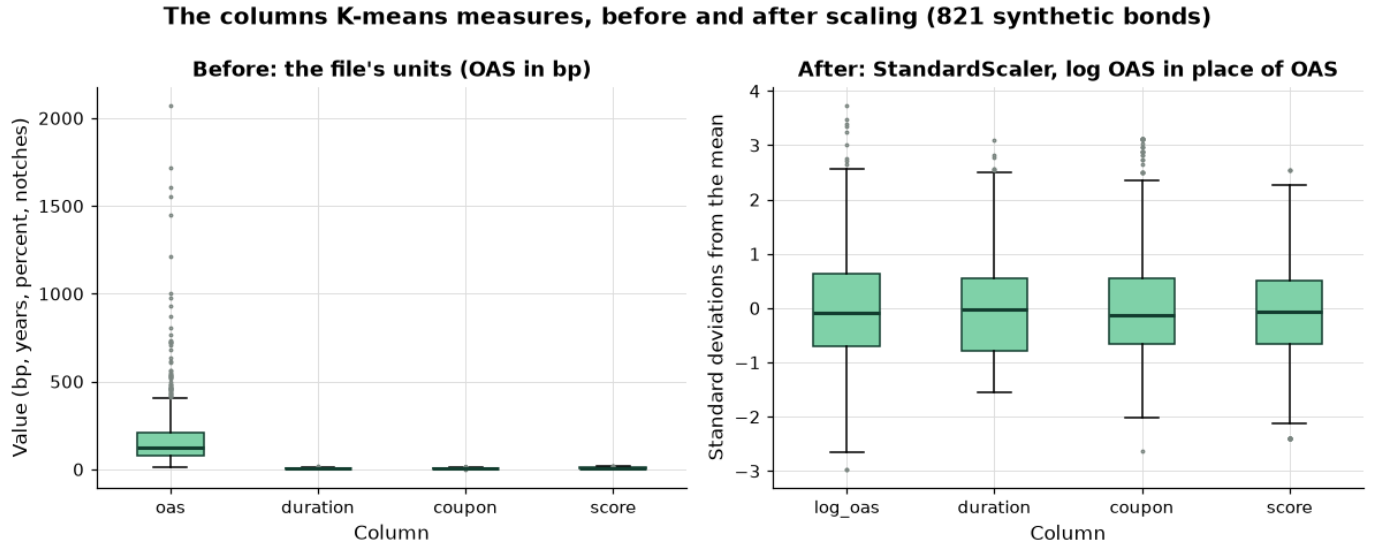
sd above the mean: widest bond, OAS in bp

3.74

sd above the mean: in logs

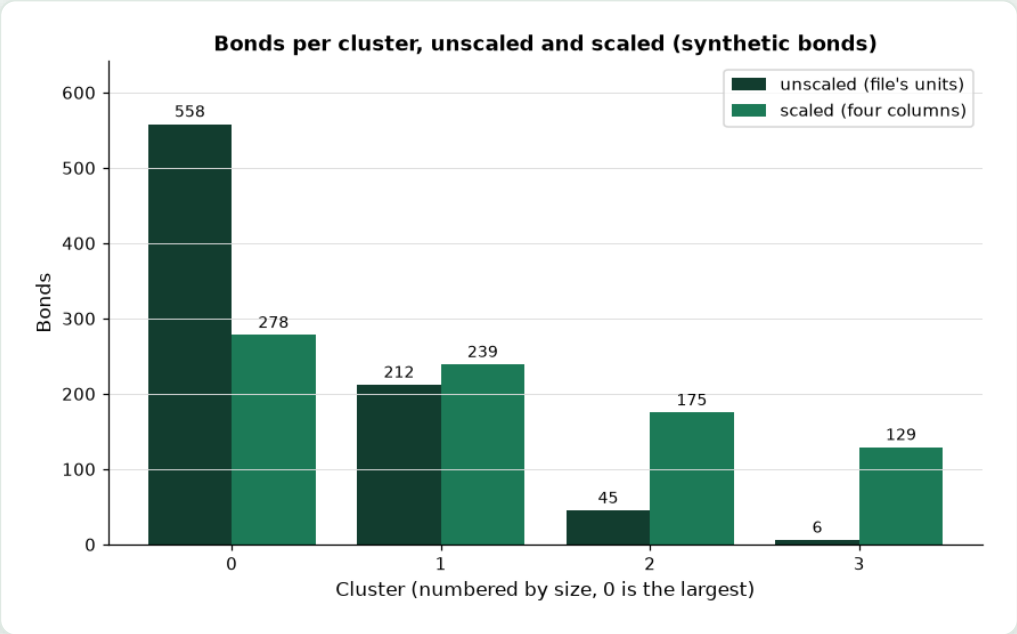
One axis, two scales

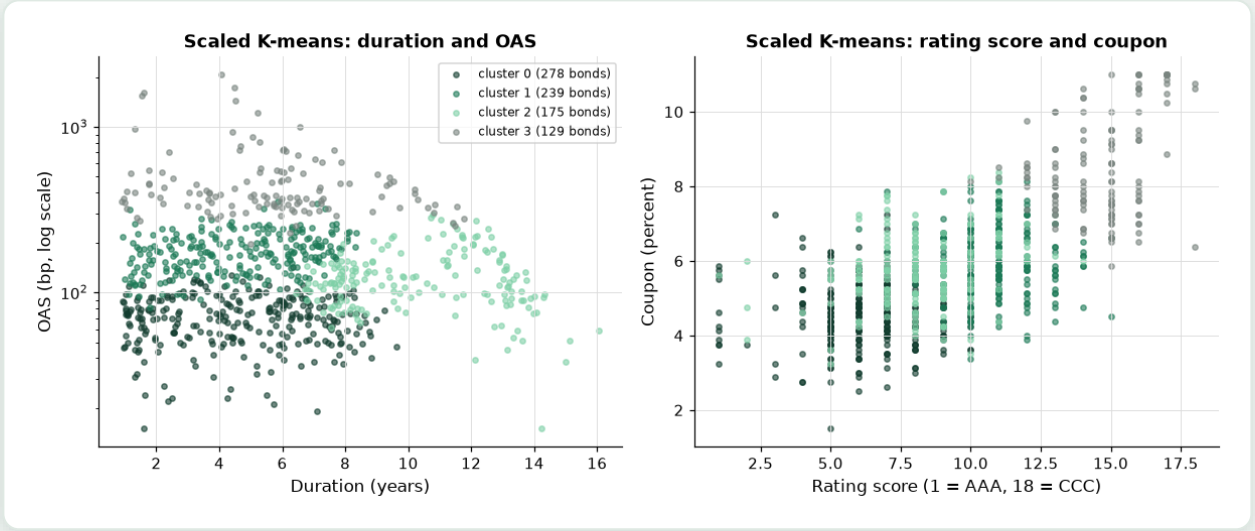
Before: OAS's middle half 79 to 210 bp, widest 2,074 bp; the other three columns stay at or below 18. After: four boxes around 0.



Scaling fixes the units, not the shape: the clusters use log OAS, as days 4 and 5 did.

Scaled, then K-means





Sizes 278, 239, 175, 129, inertia 1141.83. The ranges overlap in every column: each cluster is a region of four columns.

How the elbow and the silhouette work

The problem	K-means needs k before it starts, and the data does not give one
Inertia	Falls every time k rises, so the smallest inertia cannot choose k
The elbow	The k after which an extra cluster stops lowering the inertia by much
The silhouette	Per row, from -1 to 1: how near its own cluster against the next one; the average scores k
What comes out	An inertia and an average silhouette for each k, read side by side

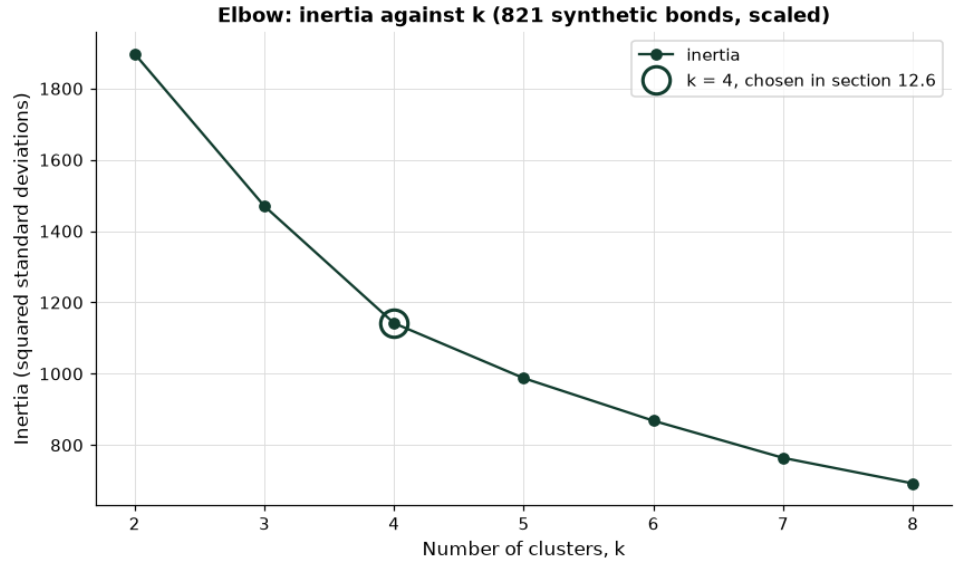
The elbow and the silhouette

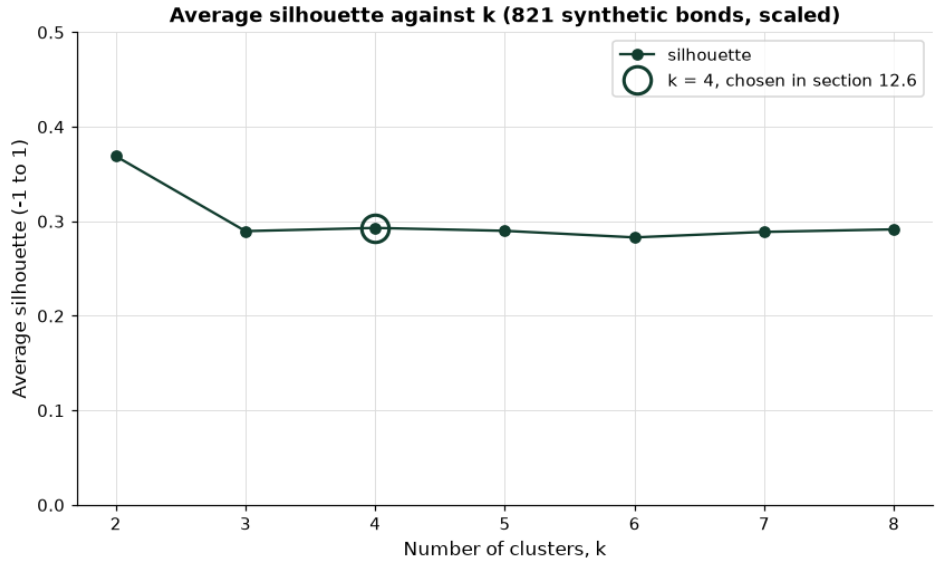
Inertia: always falls as k rises

The elbow is where one more cluster stops buying much.

Silhouette = (b - a) / max(a, b)

a: own cluster; b: nearest other. First bond: a 1.4057, b 1.6378, silhouette 0.1417. All bonds: 0.2930. elbow_table and three tests: 50 passed.

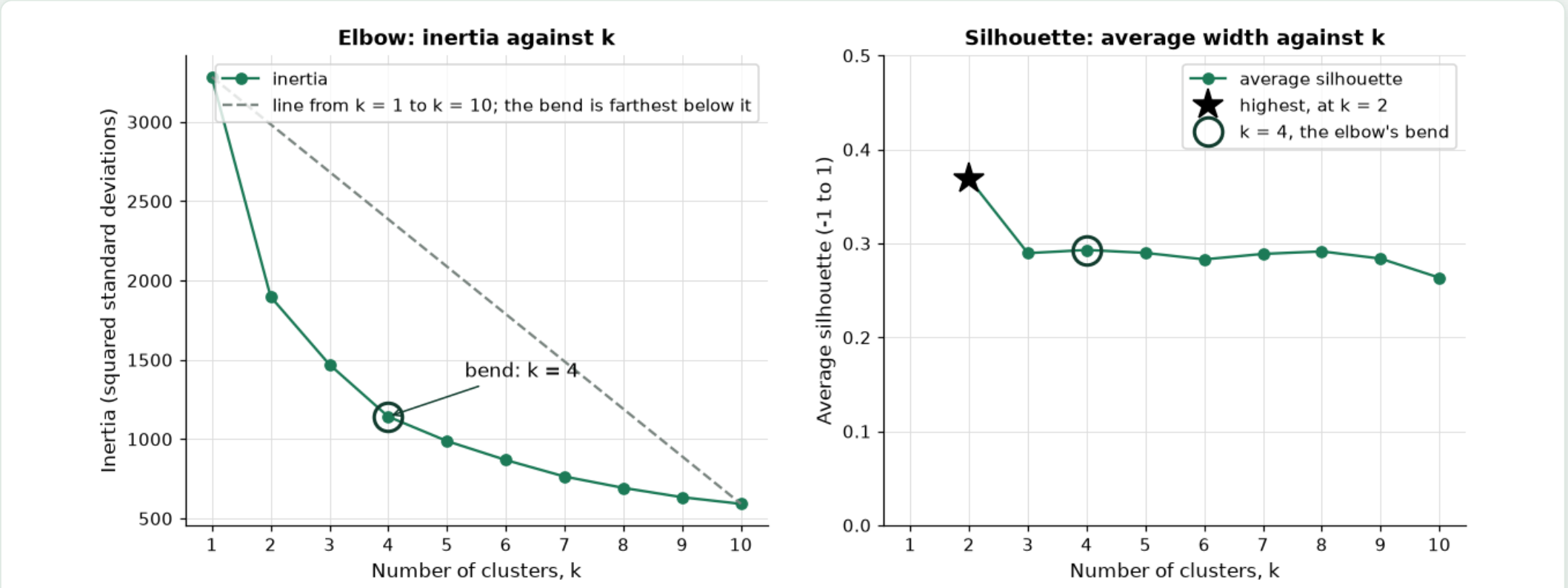




How k is chosen

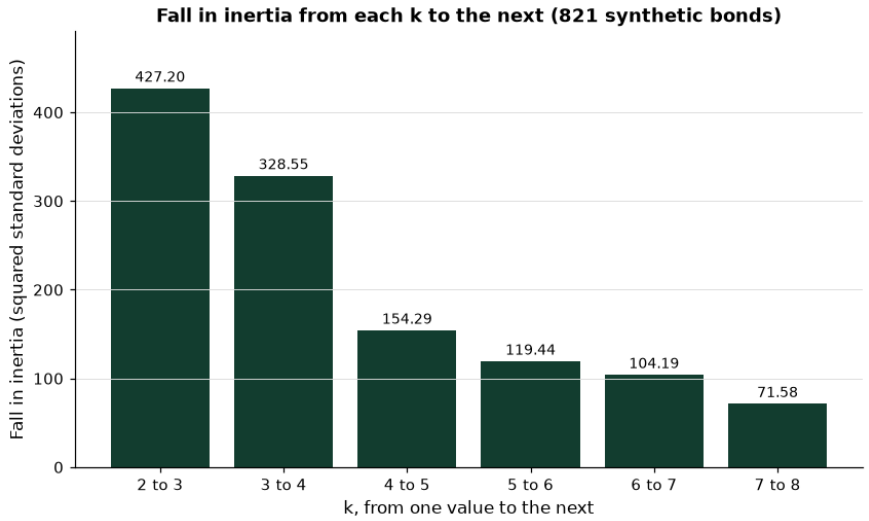
Two charts	The elbow on the left, the silhouette on the right, the same k on both axes
The bend rule	Scale k and inertia to 0 to 1; the bend is the k farthest below the line from the first point to the last
When they agree	The choice is easy
When they differ	Silhouette diagrams show thin clusters, negative widths, and uneven sizes
What comes out	A marked bend, a best silhouette, and a judgement you write down

The bend, beside the silhouette



The rule puts the bend at k = 4; the silhouette is highest at k = 2 (0.3689) and flat from 3 to 10 (0.2635 to 0.2930). They disagree: look at what each k separates.

Choosing k as a judgement



328.55 into k = 4, then 154.29 into k = 5: the fall halves.

What k = 2 separates

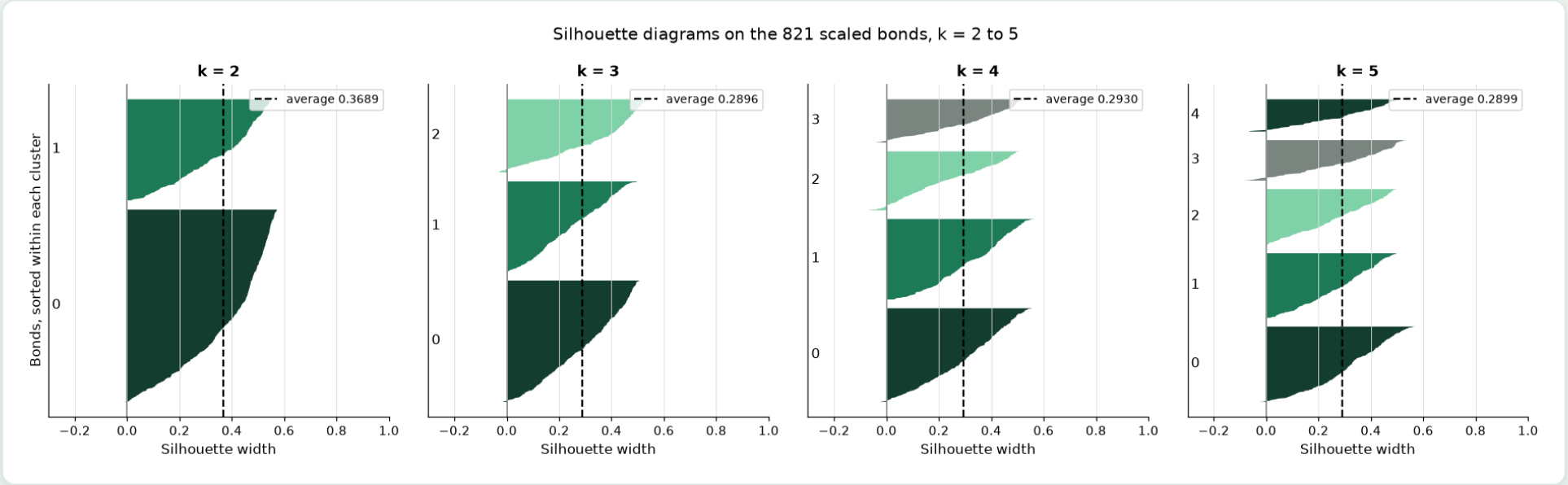
538 and 283 bonds: OAS 96.22 against 336.60 bp, score 7.34 against 12.88, duration 6.00 against 5.95 years. Duration plays no part; k = 4 adds a duration group.

The course takes k = 4

The curve bends there and its silhouette is the highest above 2. k = 2 or 5 could be argued as well.

The charts support a choice. Neither discovers a true number of clusters.

Silhouette diagrams



k	Average	Bonds below 0	Smallest cluster	Lowest cluster average	Highest cluster average
2	0.3689	1	283	0.3492	0.3792
3	0.2896	14	211	0.2502	0.3262
4	0.2930	24	129	0.2335	0.3238
5	0.2899	20	99	0.2712	0.3160

Clusters described, never ranked

Allowed names

cluster, cluster 2

a group of similar rows in an invented dataset

described by counts and averages

Never called

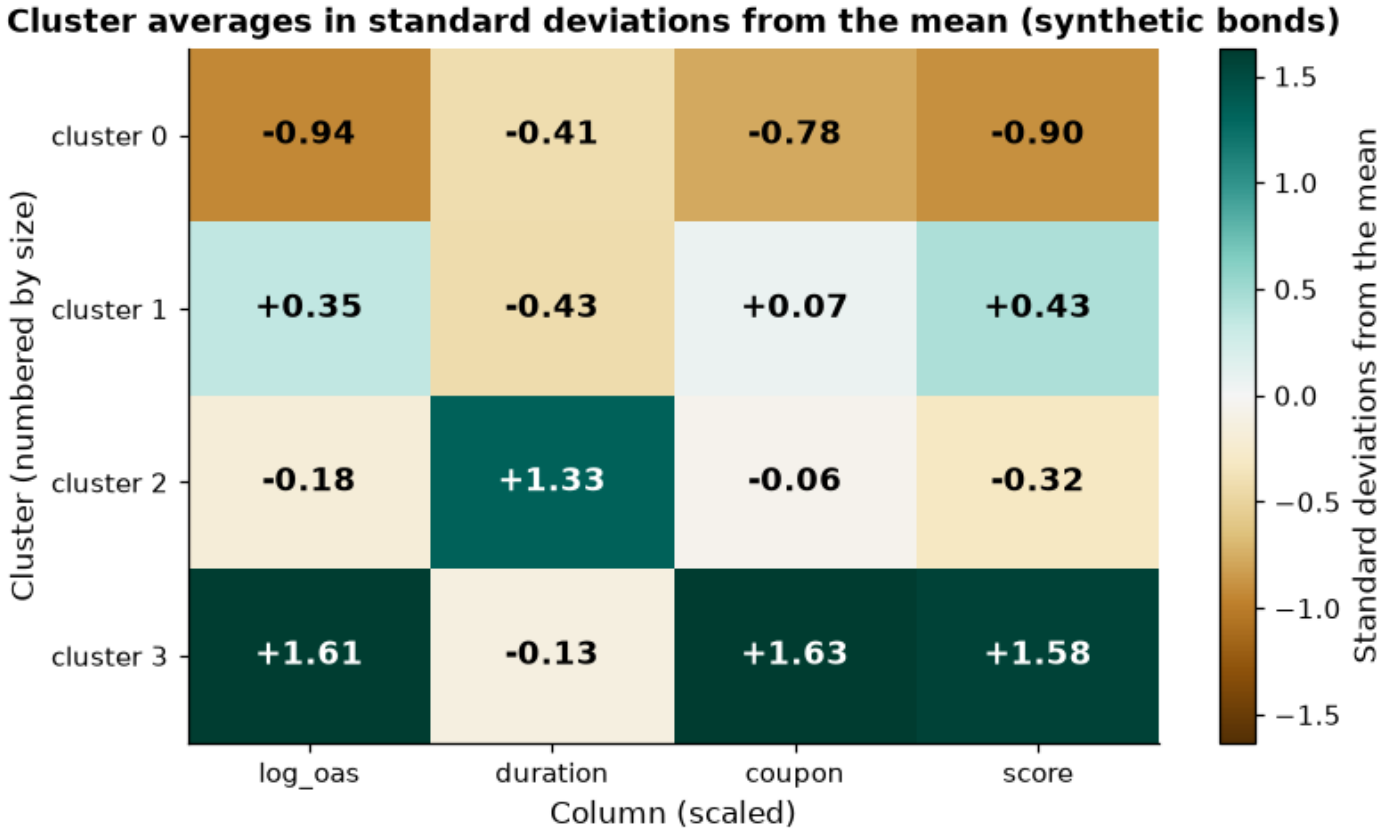
a recommendation, a ranking, a quality label

good, bad, safe, risky

cheap, rich; a peer group of real issuers

Cluster	Bonds	OAS (bp)	Duration (yrs)	Coupon (%)	Score	IG (%)
0	278	70.8	4.65	4.57	6.16	100.0
1	239	178.5	4.59	5.96	10.74	45.6
2	175	126.3	10.32	5.75	8.14	86.9
3	129	485.0	5.55	8.55	14.66	0.8

Cluster averages on one scale

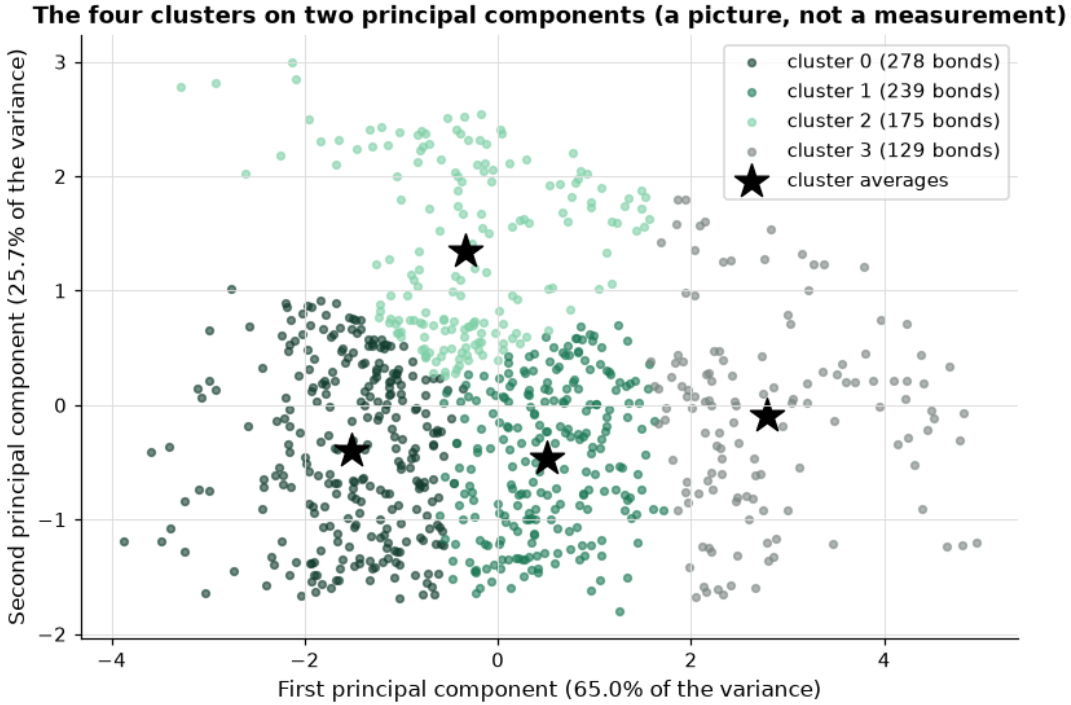


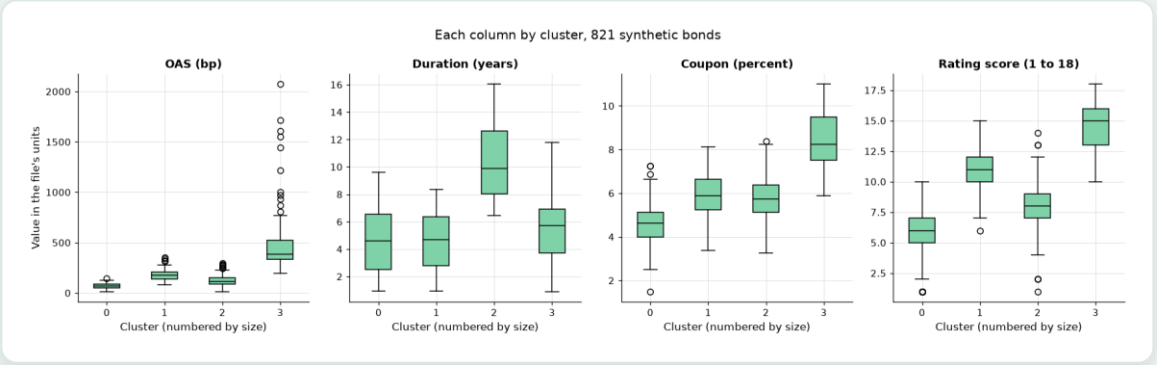
Read across a row

Cluster 2: +1.33 on duration, near 0 elsewhere. Cluster 3: +1.58 to +1.63 on log OAS, coupon, and score. Cluster 0: below the average bond on all four.

Distance from the average bond, either way. A description, not a ranking.

Clusters on two components, and their spread



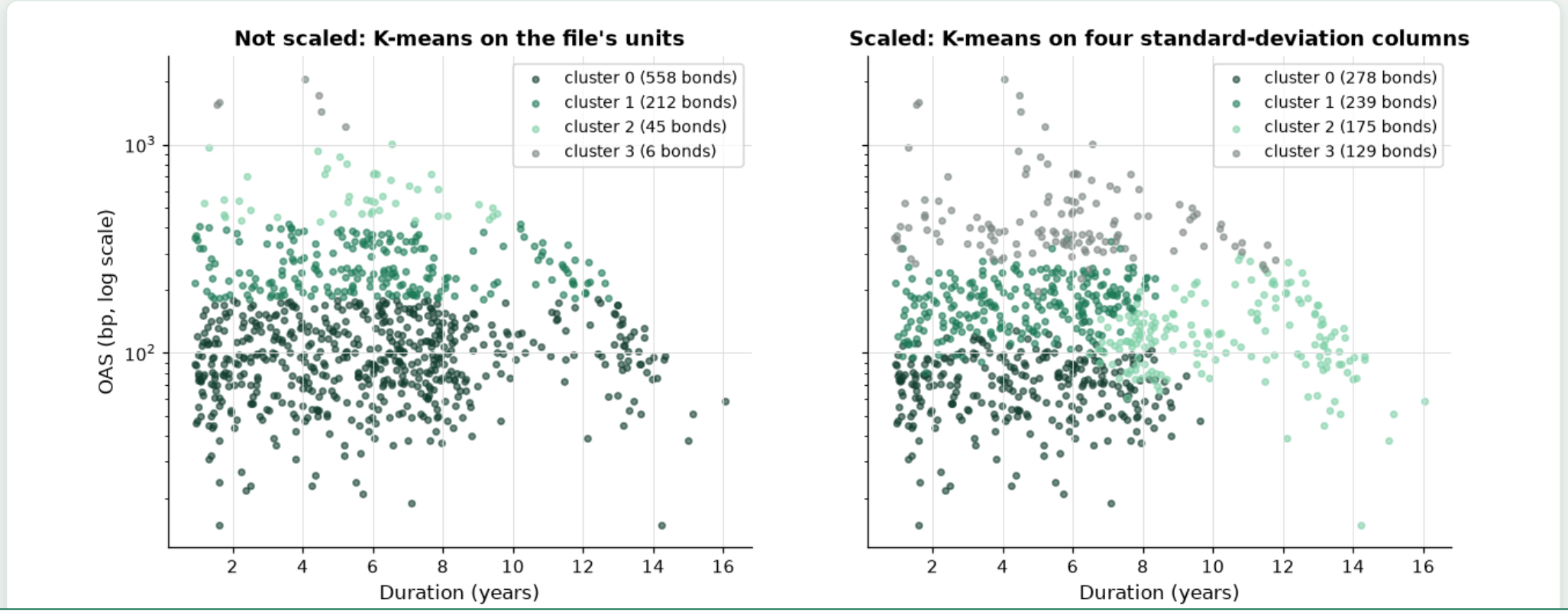


Two components keep 90.7%

65.0% and 25.7% of the variance. The first mixes log OAS, coupon, and score; the second is nearly all duration. A picture, not a measurement.

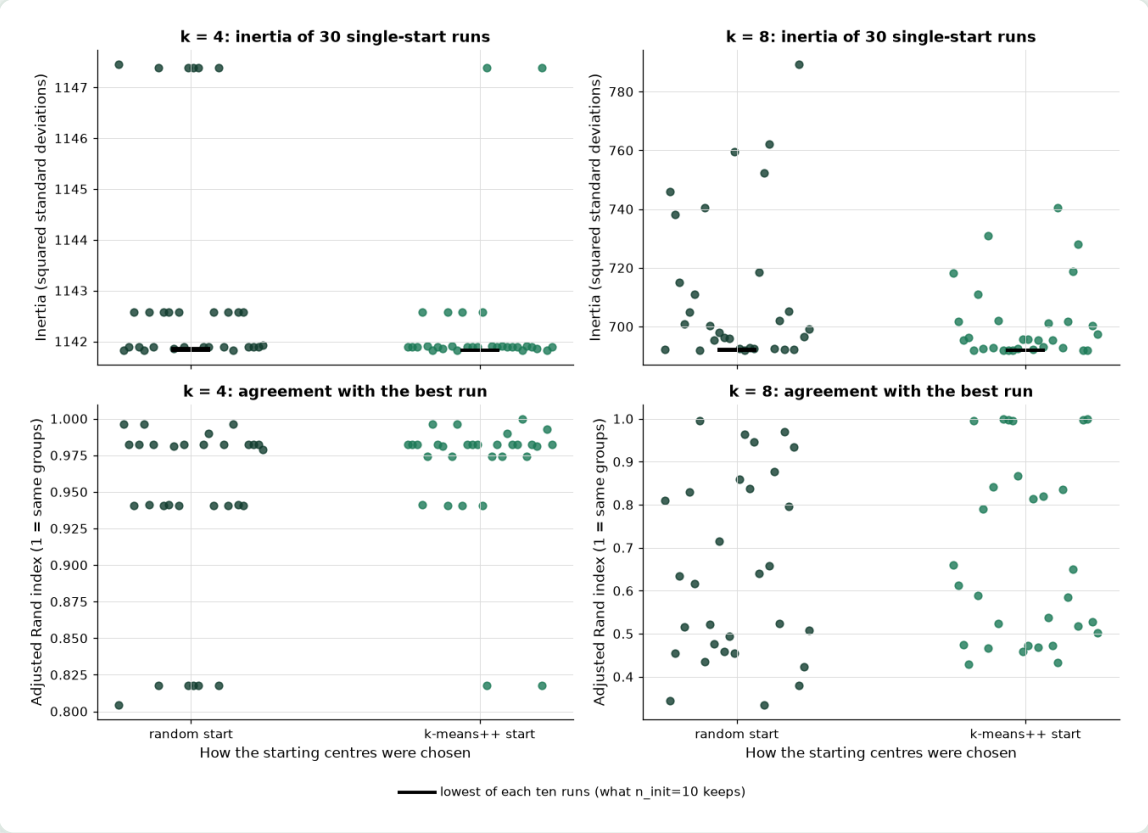
The averages say where a cluster sits; the boxes say how much it overlaps its neighbours.

Habit 1: scale first



Unscaled: bands of OAS across every duration (558, 212, 45, 6 bonds). Scaled: cut by duration and spread together (278, 239, 175, 129 bonds).

Habits 2 and 3: several starts, and do the groups hold



Run set	Median	Highest	Lowest agree
k = 4, random start	1142.25	1147.46	0.80
k = 4, k-means++ start	1141.90	1147.39	0.82
k = 8, random start	699.67	789.37	0.33
k = 8, k-means++ start	695.51	740.37	0.43

Write n_init out; never 1

At k = 4 every start lands near the best. At k = 8 single starts differ widely; the lowest of ten reaches the best. Agreement is the adjusted Rand index against the best run: 1 means the same groups.

The five habits

1. Scale first	the unscaled and scaled clusters side by side	OAS alone made 99.9 percent of the squared distance between the first two bonds; unscaled, the four clusters were four bands of OAS
2. Several starts	the lowest inertia of ten starts, n_init=10	at k = 8 a single start ended up to 98 above the best inertia; the lowest of ten reached it
3. Do the groups hold?	the adjusted Rand index against the best run	lowest agreement 0.80 at k = 4 and 0.33 at k = 8
4. Check the sizes	bonds per cluster; thin bands in the silhouette diagram	278, 239, 175, 129 bonds; the smallest holds 15.7 percent of the file
5. Choose k with evidence, describe in units	the bend, the silhouette, the diagrams, then the profile	the bend at k = 4, the highest silhouette at k = 2, four groups described by counts, averages, and boxes

Excel side by side

Excel side by side (tested in Excel)

`=STANDARDIZE(A2,AVERAGE(A$2:A$822),STDEV.P(A$2:A$822))`	StandardScaler, within 1e-9
`=STANDARDIZE(..., STDEV.S(...))`	smaller by 1.00061: not a match
`=SQRT(SUMXMY2($G2:$J2,$T$1:$W$1))`	distance to centre 0
`=MATCH(MIN(L2:O2),L2:O2,0)-1`	nearest centre: 821 of 821
`=AVERAGEIF($P$2:$P$822,$S7,G$2:G$822)`	update step: the averages
`=COUNTIF($P$2:$P$822,$S7)`	sizes 278, 239, 175, 129

Excel has no K-means, silhouette, or elbow function. It can check every step of one round.

Save the day: git show HEAD:results.csv

git show HEAD:results.csv (the last commit)

```
model,features,k,rows,inertia,silhouette
kmeans_scaled,log_oas duration coupon score,4,821,1141.8300,0.2930
```

The working file, after adding k = 2

```
model,features,k,rows,inertia,silhouette
kmeans_scaled,log_oas duration coupon score,4,821,1141.8300,0.2930
kmeans_scaled,log_oas duration coupon score,2,821,1897.5800,0.3689
```

HEAD:file is that file inside the last commit: what the log said before the change.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.

1

Docstring

Which rows fit, which rows report, and the units: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: K-means on raw OAS in bp and duration in years, read as "duration groups". Means [179.08, 5.98]: nothing scaled.

Code that runs is not the same as code that is right.

What you can do now

- No target: clustering ends in a judgement
- Unscaled, the biggest unit decides the distance
- K-means: start, assign, update, repeat
- Elbow and silhouette support a choice of k
- Clusters described by counts and averages, never ranked

Week 3, Day 3 Hierarchical clustering and cluster profiling: the 150 issuers